Candidates' Name .....

Signature: .....

Random No.					Personal No.		

(Do not write your school / Center name or Number anywhere)

Agriculture

P515/2

Theory
July / Aug, 2025
3hours

KAMTEC EXAMINATIONS BOARD

Uganda Advanced Certificate of Education
PRINCIPLES AND PRACTICES OF AGRICULTURE
THEORY
Paper 2
3Hours

INSTRUCTIONS TO CANDIDATES:
This paper consists of sections A, B, C, D and E.

Answer **question 1** *in section* **A** *and* ***four*** *other questions, selecting* **one** *from each of sections* ***B, C, D and E.***

Any additional question(s) attempted will **not** *be marked.*

For Examiners' Use Only			
Section	**Question**	**Marks**	**Examiner's signature and No.**

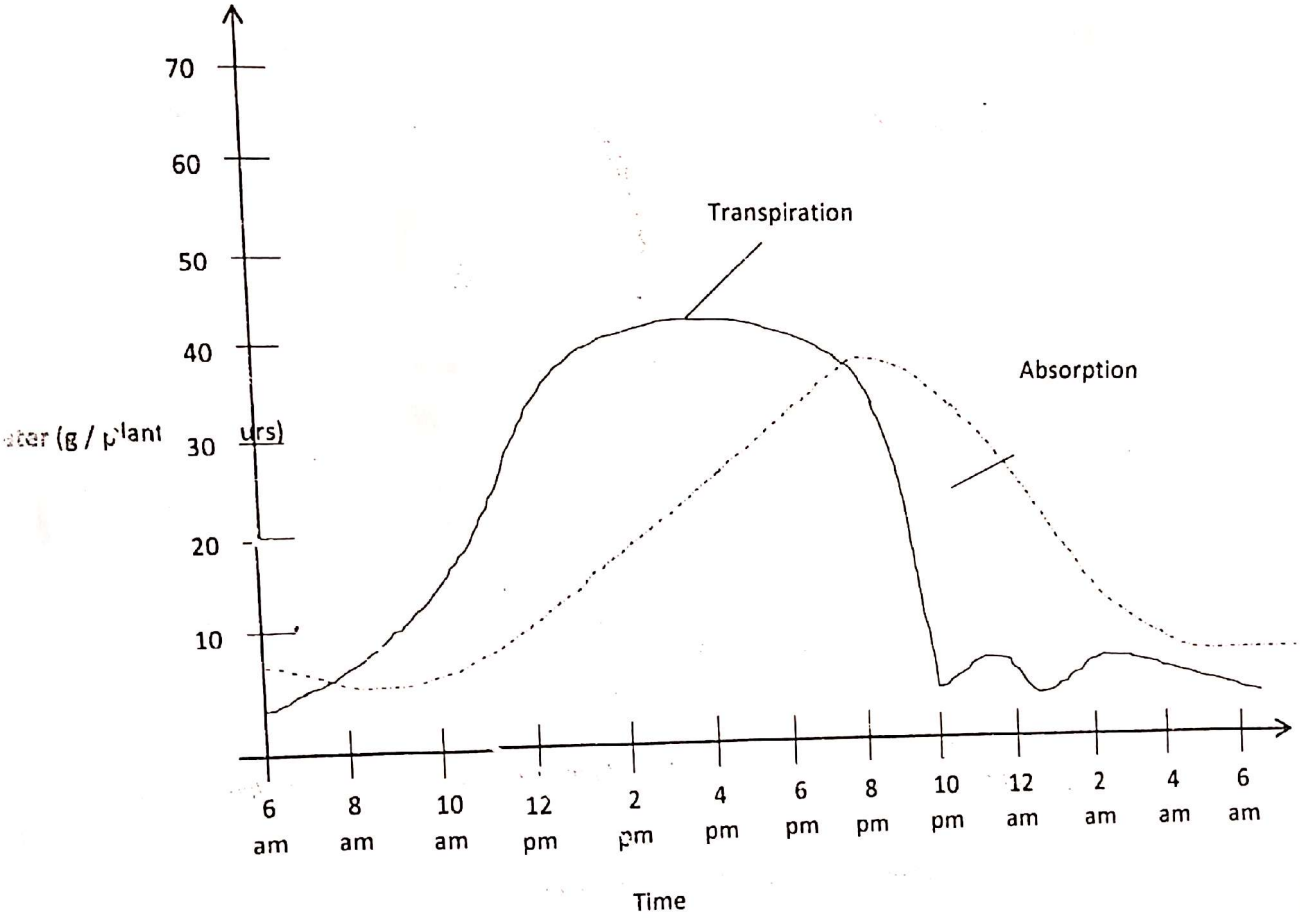
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SECTION A: COMPULSORY QUESTION (20 MARKS)

Question 1 is compulsory

1. The information below is about the rate of transpiration and water absorption by a sun flower plant (*Helianthus annuus)* during a hot summer day and the following night. Study the information and answer the questions that follow:



(a) (i) Define transpiration. *(01 mark)*

(ii) Describe the nature of transpiration and absorption for the plant over time. *(07 marks)*

(b) Mention the time of the day at which transpiration is equal to absorption. *(01 mark)*

(c) Account for the rate of transpiration between: *(03 marks)*

(i) 6:00am – 10:00 am

(ii) 4:00pm – 10:00pm

(d) (i) What is the relationship between transpiration and absorption between

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8:00am and 10:00am?

(iii) Give two factors that affect the rate of water absorption besides transpiration. *(02 marks)*

(iv) What is the significance of transpiration in the crop of sunflower? *(04 marks)*

(02 marks)

SECTION B
CROP PRODUCTION (20 MARKS)

2. (a) What is organic farming? *(02 marks)*

(b) Explain the various agriculture practices enhance organic farming in Uganda. *(12 marks)*

(c) Outline the benefits of organic agriculture. *(06 marks)*

3. (a) Explain the measures that farmers can take to conserve soil moisture. *(10 marks)*

(b) What causes fluctuations of organic matter levels in soils? *(04 marks)*

(c) How does poor soil drainage affect soil fertility and productivity? *(06 marks)*

SECTION C
ANIMAL PRODUCTION (20 MARKS)

4. (a) What are the factors that may encourage hive colonization by bees? *(05 marks)*

(b) Explain the practices that can be carried out to improve the level of honey production in bees. *(10 marks)*

(c) Outline the advantages of apiculture over other enterprises. *(05 marks)*

5. (a) Explain why artificial insemination technology should be encouraged in livestock farming communities of Uganda. *(10 marks)*

(b) What are the constraints to the adoption and development of artificial insemination technology in Uganda? *(10 marks)*

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SECTION D
ECONOMICS (20 MARKS)

6. (a) Explain the steps that government has taken to increase women participation in agriculture. ***(12 marks)***

(b) Give the challenges that government faces while implementing policies in agriculture. ***(08 marks)***

7. (a) Explain the roles that research stations play in the development of agriculture. ***(12 marks)***

(b) Outline the challenges faced by research centres in executing their mandate. ***(08 marks)***

SECTION E
FARM STRUCTURES AND MACHINES (20 MARKS)

8. (a) Describe the procedure followed to construct a simple fish pond ***(12 marks)***

(b) Outline the steps taken to maintain a fish pond in a good state for high fish yields. ***(08 marks)***

9. (a) Explain the work done by various simple machines in farm work. ***(12 marks)***

(b) What management practices can prolong the life span of simple machines? ***(08 marks)***

END

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